

Human Development Index (HDI) Prediction & Socioeconomic Sandbox

Project Overview

The **Human Development Index (HDI) Prediction & Socioeconomic Sandbox** is an interactive full-stack web application developed to simulate, analyze, and predict Human Development Index (HDI) values using socioeconomic indicators. The platform enables analysts, researchers, students, and policymakers to understand how variations in health, education, and income influence a country's human development. By combining the official United Nations Development Programme (UNDP) HDI calculation methodology with advanced machine learning models and Google Gemini AI, the application provides accurate predictions, interactive visualizations, and AI-generated policy recommendations. The project serves as both an educational platform for learning socioeconomic analysis and a decision-support tool for research and policy planning.

Project Objectives

The primary objective of this project is to build an intelligent platform capable of predicting Human Development Index values based on user-defined socioeconomic inputs. The system allows users to simulate different development scenarios, compare the performance of multiple machine learning models, and visualize prediction results through interactive charts. Another objective is to generate AI-powered policy recommendations that help researchers and policymakers identify development challenges and recommend practical improvement strategies. The platform is also intended to provide an easy-to-use environment where users can understand the relationship between education, healthcare, income, and overall human development.

Key Features

The application offers several powerful features that make socioeconomic analysis simple, interactive, and informative.

Important Features

- Interactive HDI simulation
- Machine Learning-based HDI prediction
- Multiple ML model comparison
- AI-generated policy recommendations
- Interactive charts and visualizations

- Scenario-based socioeconomic sandbox
- Responsive user interface
- Offline fallback support

Users can modify socioeconomic variables such as life expectancy, expected years of schooling, mean years of schooling, and Gross National Income (GNI) per capita. As soon as these values are changed, the system recalculates the HDI and generates updated predictions. The application also compares multiple machine learning algorithms and visualizes their performance using graphs and charts. Google Gemini AI further analyzes the prediction results and automatically generates detailed policy reports highlighting strengths, weaknesses, and recommendations for future development.

System Workflow

The workflow begins when the user selects a country and enters the required socioeconomic indicators. The application then calculates the HDI components according to the UNDP methodology, executes the selected machine learning model to predict the HDI, generates interactive visualizations, and finally creates an AI-powered policy report. This workflow provides users with numerical predictions, graphical insights, and intelligent recommendations in a single integrated platform.

Project Architecture

The application follows a modern client-server architecture. The frontend is developed using React and Tailwind CSS, providing a responsive and interactive user experience. The backend is built with Express.js running on Node.js, which processes user requests, performs HDI calculations, executes machine learning models, and communicates with Google Gemini AI. After processing the user's inputs, the backend returns prediction results, visualization data, and AI-generated reports to the frontend for presentation through an interactive dashboard.

How to Run the Project

Running the **Human Development Index (HDI) Prediction & Socioeconomic Sandbox** project locally is straightforward. Before starting, ensure that **Node.js version 18 or later** and **npm** are installed on your system. Open the project folder in **Visual Studio Code** by selecting **File → Open Folder** and choosing the project's root directory. Once the project is opened, launch the integrated terminal by pressing **Ctrl + ** (Windows/Linux) or **Cmd + (Mac)**.

Step 1: Verify Node.js and npm Installation

Before installing dependencies, verify that Node.js and npm are installed correctly by executing the following commands:

```
node -v
```

```
npm -v
```

If both commands return version numbers, your environment is ready for the next step.

Step 2: Install Project Dependencies

Install all required frontend and backend dependencies by running:

```
npm install
```

This command downloads and installs every package required by the application.

Step 3: Configure Environment Variables

Create a file named **.env** in the project's root directory (the same location as the **package.json** file). Add the following environment variables:

```
GEMINI_API_KEY="your_actual_gemini_api_key_here"
```

```
APP_URL="http://localhost:3000"
```

Replace **your_actual_gemini_api_key_here** with your valid Google Gemini API key.

Step 4: Start the Development Server

Run the following command to launch both the React frontend and Express backend:

```
npm run dev
```

After the application starts successfully, the terminal will display:

```
[HDI Server] Active and listening on http://localhost:3000
```

Open your preferred web browser and navigate to:

```
http://localhost:3000
```

The application dashboard will load, allowing you to simulate HDI scenarios, compare machine learning models, generate interactive visualizations, and receive AI-powered policy recommendations.

Step 5: Build the Application for Production

To create an optimized production build, execute:

```
npm run build
```

This command compiles and bundles both the frontend and backend into production-ready files.

Step 6: Run the Production Server

After the build process is complete, start the production server using:

```
npm start
```

The optimized version of the application is now ready for deployment or local production testing.

Prerequisites

Before running the project, ensure the following software and resources are available:

- Node.js (Version 18 or later)
- npm (included with Node.js)
- Visual Studio Code
- Google Gemini API Key
- Internet connection (required for AI-generated reports)
- Modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari)

Following these steps will successfully launch the application and enable all HDI prediction, visualization, machine learning, and AI reporting features.

HDI Calculation Methodology

The application follows the official Human Development Index methodology established by the **United Nations Development Programme (UNDP)**. The HDI is calculated using three major dimensions of human development: Health, Education, and Income.

The **Life Expectancy Index (LEI)** evaluates health performance using life expectancy values ranging from 20 to 85 years. Values outside this range are automatically adjusted to remain within the accepted limits.

The **Education Index (EI)** is calculated using two educational indicators: Expected Years of Schooling and Mean Years of Schooling. These indicators collectively represent the educational achievement of a country.

The **Income Index (II)** measures economic development using Gross National Income (GNI) per capita. Since income contributes diminishing improvements to quality of life, the application applies a logarithmic transformation while limiting values between **\$100** and **\$75,000**.

The final Human Development Index is calculated as the geometric mean of the Life Expectancy Index, Education Index, and Income Index.

Database Structure

The application organizes information using eight primary entities that resemble a relational database.

Primary Entities

- User
- Session
- Country
- HDI Input Data
- Dataset
- Machine Learning Model
- HDI Prediction
- Visualization Report

Each entity plays a specific role within the system. The **User** entity stores analyst information, while **Session** records user activity. **Country** maintains baseline socioeconomic data, and **HDI Input Data** stores user simulation values. **Dataset** manages machine learning training datasets, **Machine Learning Model** stores algorithm configurations and evaluation metrics, **HDI Prediction** stores prediction outputs, and **Visualization Report** contains AI-generated policy reports and visualization data.

Technology Stack

The application utilizes a modern full-stack technology stack.

Frontend Technologies

- React 19
- Tailwind CSS 4
- Recharts
- Motion (Motion/React)
- Lucide React
- React Markdown

Backend Technologies

- Node.js
- Express 4
- Vite
- Esbuild
- TSX

The frontend provides a responsive interface with interactive visualizations, while the backend manages API requests, HDI calculations, machine learning predictions, and AI integration.

Artificial Intelligence Integration

The application integrates Google's **@google/genai (Version 2.4.0)** SDK to generate intelligent policy recommendations using the **Gemini 3.5 Flash** model. Once prediction results are generated, the application sends the socioeconomic indicators and calculated HDI values to Gemini AI, which produces a structured policy report describing the country's current development status, identifying major challenges, and recommending practical improvement strategies.

Dual-Model Fallback System

To ensure uninterrupted service availability, the application implements a robust three-stage fallback mechanism.

Fallback Strategy

- Primary AI Model – Gemini 3.5 Flash
- Secondary AI Model – Gemini 3.1 Flash Lite
- Local Rule-Based Report Generator

If the primary AI model becomes unavailable due to high server demand, the application automatically switches to the secondary model. If both online models fail or an API key is unavailable, the application activates a built-in rule-based engine that generates a complete Markdown report locally, ensuring continuous operation even without internet connectivity.

Advantages

The project provides numerous advantages for researchers, students, analysts, and policymakers.

Major Advantages

- Interactive HDI prediction
 - Real-time socioeconomic simulation
 - Multiple machine learning model comparison
 - AI-generated policy recommendations
 - Modern responsive user interface
 - Interactive visualizations
 - Offline report generation
 - Easy deployment and scalability
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Future Enhancements

The project can be expanded by integrating live datasets from organizations such as the United Nations and the World Bank. Future improvements also include secure user authentication, PDF and Excel export functionality, multilingual support, LSTM-based forecasting models, interactive world maps, cloud deployment, and collaborative research features. These enhancements will significantly improve the accuracy, usability, and scalability of the platform.

Conclusion

The **Human Development Index (HDI) Prediction & Socioeconomic Sandbox** is a comprehensive full-stack application that combines standardized HDI calculations, machine learning prediction models, interactive visualizations, and Google Gemini AI to provide an intelligent socioeconomic analysis platform. By allowing users to simulate development scenarios, compare predictive models, and generate AI-powered policy reports, the application serves as an effective educational resource, research platform, and policy decision-support system. Its modern architecture, resilient fallback mechanisms, and advanced technology stack ensure reliable performance while delivering valuable insights into human development and socioeconomic planning.